

NMAM INSTITUTE OF TECHNOLOGY, NITTE
 (An Autonomous Institution affiliated to VTU, Belagavi)
Eighth Semester B.E. (Credit System) Degree Examinations
 May - June 2023

19EC8X18 - CONSUMER ELECTRONICS

Duration: 3 Hours

Max. Marks: 100

- Note: 1) Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.
 2) Any missing data maybe suitably assumed.

Unit – I		Marks	BT*	CO*	PO*
1.	a) Explain the working of capacitive microphone with relevant diagram.	8	L*2	1	1
	b) What are baffles? What is the necessity of a baffle in speakers?	4	L2	1	1
	c) Explain the Moving Coil Electromagnet Cone Speaker with relevant diagram.	8	L2	1	1
2.	a) Explain the construction and working of ribbon microphone with necessary diagram.	8	L2	1	1
	b) What is baffle? Explain the necessity of baffles in cone type of speakers.	7	L2	1	1
	c) Explain traffic control workflow using CCTV system.	5	L2	2	1
3.	a) Explain the constructional features of dynamic cone speaker with necessary diagram.	7	L2	1	1
	b) Describe the different types of LED display.	5	L2	2	1
	c) Explain the working principle of LCD with relevant diagrams.	8	L2	2	1
Unit – II					
4.	a) Explain the principles of Uniselectors used in switching systems.	8	L2	4	1
	b) Explain Pulse code modulation.	7	L2	3	1
	c) Mention the merits of DTH service.	5	L2	3	1
5.	a) What is HiFi system? Discuss the various distortions of HiFi system.	6	L2	4	1
	b) Explain the working of disc recording and reproduction with relevant diagrams.	7	L2	3	1
	c) Explain different Methods of Broadcasting Color TV Signals.	7	L2	4	1
6.	a) Explain the working principles of superheterodyne receiver with relevant diagram.	7	L2	4	1
	b) Explain the schematic arrangement of cable TV network.	7	L2	4	1
	c) Write a short note on satellite system by describing the various components of earth station and satellite station.	6	L2	4	1
Unit – III					
7.	a) Explain the basic operation of fax machine with relevant block diagram.	8	L2	5	1
	b) Explain the process of electronic ignition of automobiles with relevant block diagram. What are the advantages of electronic ignition system over mechanical ignition system?	8	L2	5	1
	c) Discuss the importance of electronics in healthcare systems.	4	L2	5	1
8.	a) Explain the working principle of Microwave Oven with a relevant block diagram.	7	L2	5	1
	b) Describe the various components of robots.	6	L2	5	1
	c) What is nanotechnology? Write a short note on carbon nanotubes.	7	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
